CHEMISTRY

BY

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RESEARCH AND THEORY
THE ATOM AS A PART, VALENCE AS THE RULE, THE BOND LENGTH AS A FACT

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For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

The subject is the building block and the rule. An atom is a point carrying values: a count of protons, a count of valence electrons, a radius, an electronegativity. A molecule or a solid is a set of such points in space. What tells that set from a random packing of the same atoms is one rule, valence, and valence fixes a bond to a length before anyone measures it. The method this research paper is written for reads any domain as points carrying values and asks one question of the arrangement: how far it sits from the most disordered arrangement of its own parts. Chemistry is that question with the atom as the part.

What is quoted and what is new. The building blocks and the rules in the first four chapters are the settled content of general chemistry: the periodic law, the shell counts, the octet, the electronegativity scale, the covalent and van der Waals radii, and the bond lengths tabulated from crystallography and gas-phase spectroscopy. Nothing about them is claimed here. What this research paper adds is their placement in the method, and the predictions that placement makes. The atom is a representation, the octet is a screening condition, the disordered packing is a reference, and the bond length is a test oracle of the strongest kind, a fact and not a consensus.

Nothing in this research paper is measured here. The method has representations for a protein and for a crystal, and both already rest on chemistry: a backbone repeats at three because three bond lengths are fixed, and a cell edge is a bond geometry tiled. It has no representation for a molecule itself. This research paper states what such a reading would find, with the predictions on the page before the reading exists.

Why no set here is bounded. A bound put on a quantity by judgment makes the reading report the bound. Chemistry invites that error in a sharp form, because a bond either exists or it does not and a tolerance is the obvious way to decide. This research paper keeps to the opposite discipline: a departure is measured against a background drawn from the atoms themselves, and neither threshold, cutoff radius nor bounded element set is chosen by judgment. The question asked is always how far, never whether.

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1

THE ATOM IS A POINT CARRYING VALUES

1.1 ONE PART, SEVERAL MAGNITUDES

A part in chemistry is an atom, and an atom is not one number. It carries a count of protons Z, which names the element and orders the whole set; a count of valence electrons, the outer shell's occupancy; a radius, in two readings, the covalent one at which it bonds and the van der Waals one at which it merely touches; and an electronegativity [6], the pull it puts on a shared pair. Written together these are a vector, and every reading in this research paper is a magnitude taken from it or a distance taken between two of them. The method reads any domain as points carrying values, and the atom is that description with the values fixed by the element.

Two atoms in a molecule are two such points at two positions, and the bond between them is the vector from one to the other. Its magnitude is the bond length. The representation has two parts: positions carrying element vectors, and the differences between positions carrying lengths. Nothing about it is particular to chemistry until the values are filled in. Hence the representation is the only part of the method that knows an atom exists.

1.2 THE ORDER IS A REAL READING, NOT A NUMBERING

The elements are ordered by Z [4], and that order is not a label chosen for convenience. Read a property along it, the atomic radius or the first ionization energy, and the sequence does not wander. It rises and falls with a period, and the period is the shell filling: two, then eight, then eight, then eighteen, then eighteen, then thirty-two. The periodic law [3] is the statement that a property of the atom is a periodic function of Z, and in the method's terms it is the first departure chemistry offers. A property read along Z agrees with itself at the shell boundaries and disagrees with a shuffle of the same values.

The shell counts are themselves a departure from a flat expectation. A shell n holds $2n^2$ electrons because that is how many distinct states the quantum numbers allow it [5]. The capacities are $2, 8, 18, 32$ and the period lengths follow from the order the sub-shells fill. This is not a count anybody chose. It is fixed before any element is weighed, the property the later chapters lean on: an answer known before the measurement and not supplied by it.

The element order, the electron set and the shell counts are the atomic structure, and this research paper reads them as given. It does not establish them. They are the data chemistry consumes, and everything this research paper adds is the bonding that sits on top of them.

1.3 THE SET IS NOT BOUNDED

There is a temptation to write chemistry as a fixed table of some number of elements. A bound put on a quantity by judgment makes the reading report the bound, and a bounded element set is that defect in a new place. The count of known elements is a fact about what has been synthesized, not about what the periodic law describes, and the law describes a sequence with no last term until a nucleus cannot hold together. A reading that fixes the set at a round number reads that number and not the chemistry.

The same holds one level down. An element is not one atom either. It is a set of isotopes differing in neutron count, weighted by abundance, and the weighting is a measured distribution and not a constant. Where a reading needs a mass it needs the distribution behind it, and a single tabulated value is a summary of that distribution with its spread deleted. The building block is a set, at every level at which it is read, and the discipline is to draw the background from the set and not to cap it.

2

VALENCE IS A NECESSARY CONDITION

2.1 THE RULE THE SCREENING ALREADY STATES

The method's screening rests on one proposition [8]: any subset of a pattern's points is a necessary condition. No selection rule built from the pattern can lose a true occurrence, and the converse fails for any proper subset. Every surviving candidate still has to be confirmed. Errors are one directional. A discrepancy is always an over-count, and it is detectable without knowing the answer.

Valence is a proposition of exactly that shape. A main-group atom in a stable molecule completes its octet [2, 1]: eight electrons in the outer shell, two for hydrogen and helium, counting shared pairs once for each atom that shares them. This is a condition on every atom at once, and a real molecule satisfies all of them. The octet is therefore a necessary condition on an arrangement, and a candidate structure that fails it at any atom is not a molecule. No true structure is ever refused by the octet, because a true structure meets it everywhere. The rule discards candidates, and that is cost, not correctness, the screening's own division carried onto chemistry unchanged.

2.2 A BOOLEAN READING AT EACH ATOM

Read one atom and its bonds and the question is boolean. Sum the electrons it owns outright and the pairs it shares, compare against the shell it is filling, and the atom either closes or it does not. Carbon with four single bonds closes at eight and is true; carbon with three is a radical and is false; carbon with five cannot be drawn at all. The reading needs no tolerance. An electron pair is present or absent, a bond is drawn or not, and the count is an integer compared against an integer.

The molecule is the conjunction. It is true when every atom closes, and the screening's intersection is the operation that reads it: each atom's condition is a set of admissible local environments, and the arrangement survives where it lies in all of them at once. This is the cascade the method runs on a text or a lattice, with the key being an atom's satisfied octet instead of a rare symbol at an offset. Nothing is ported to bring it here. The screening is written over an index set with no interpretation, and an atom is one more thing an index can hold.

2.3 WHY THE EXACT COMPARE IS IRREDUCIBLE

The octet selects candidates and does not confirm them, exactly as the proposition warns. A drawn structure that satisfies every octet can still be the wrong isomer, because the same atoms with the

same valences admit more than one arrangement: ethanol and dimethyl ether share a formula and both close every atom. The rule narrows the field to the arrangements that could be molecules and leaves the choice among them to something the rule does not see, which for a real substance is its energy and for this method is a departure measured against the disordered packing of the next chapter.

The one-directional error in chemistry's terms: a structure the octet refuses is refused rightly, because it cannot be a closed-shell molecule. A structure the octet admits is not thereby a molecule. A search that uses valence to prune loses nothing real and keeps some things that are not real, the only failure the screening permits and the reason a verification pass is never optional.

3

THE BOND LENGTH IS A TEST ORACLE

3.1 A FACT, NOT A CONSENSUS

The method keeps its outside answers apart, as test oracles, and grades against them. A family tree of languages is a reconstruction argued from cognates. Agreement with it is agreement with a scholarly consensus, and where the two disagree nothing in the method can say which is wrong. A published crystal cell edge is the other kind: somebody measured it, refereed it and wrote it down, and the instrument either recovers it or does not. The bond length is that second kind and more so. Valence fixes it whatever anyone believes. A carbon-carbon single bond is near 1.54 angstroms, a double near 1.34, a triple near 1.20, and a substance does not get a vote.

Bond lengths therefore belong apart from the family trees, because a bond length is a fact and a family tree is a consensus and the two must not be graded as the same kind of answer. The values the method consults for a protein already come from this table: the backbone bonds are near 1.46, 1.52 and 1.33 angstroms, nitrogen to alpha carbon, alpha carbon to carbon, and the peptide carbon to the next nitrogen. None of those three is measured by the instrument that reads the backbone. They are the answer it is checked against.

3.2 TWO DOMAINS ALREADY REST ON IT

The method has a representation for a protein and one for a crystal, and both stand on chemistry without naming it. The protein backbone is a chain of bonds whose lengths cycle with a period of three, because the three bond types repeat down the chain, and that period is fixed by chemistry before the chain is measured. It is a positive control with an answer not produced by the sample: a detector that returns anything other than three there is wrong, and no argument afterward can save it. The three comes from valence, and valence is chemistry.

The crystal is the same fact tiled. A cell edge is a repeat distance, and what sets the repeat is the sizes of the atoms and the lengths of the bonds between them, packed into the smallest cell that reproduces under translation. The instrument recovers 453 of 453 published edges inside one voxel [7], and the thing it is recovering is a bond geometry stacked. Chemistry is not a third domain beside the protein and the crystal. It is the layer under both, the building blocks and the one rule that fix the lengths those two readings recover.

3.3 THE FACT MAKES THE TRAP VISIBLE

A test oracle that fixes a period exactly is also what exposes a detector's own failure. A sequence of period P agrees with itself as well at $2P$ and $3P$. A detector that takes the tallest lag reports a harmonic about as often as the period. On the protein backbone chemistry says the answer is three, and a detector returning six or nine is wrong on the spot. Without the fixed answer the harmonic passes as a clean result, a confident number with a factor of two that nothing flags.

That is the argument for chemistry as a test oracle stated from the other side. It supplies an answer, and the answer is exact and external. An instrument checked against it cannot hide an error inside a tolerance. A bond length carries no spread that a detector's mistake can shelter in. It is the hardest test available because the sample can argue with it least.

4

THE DISORDERED ARRANGEMENT OF THE SAME ATOMS

4.1 THE BACKGROUND IS DRAWN, NOT ASSUMED

The question the method asks is always how far an arrangement sits from the most disordered arrangement of its own parts, and the most disordered arrangement is not a model to be posited. It is drawn from the object by deleting one property. Take the atoms a molecule holds, keep their counts exactly, and scatter their positions: the result is the least committal distribution consistent with the composition, the constrained maximum of entropy, and it sits at a single point because entropy is concave and a count constraint is linear. It carries no seed and no run-to-run variation, and a substance with no arrangement, its atoms placed at random, returns a departure of nothing.

Chemistry makes this concrete in a way a text does not. The composition is the formula, the shuffle is the same atoms at scrambled positions, and the departure is everything the bonds contribute over a gas of the same elements. A tolerance is the wrong tool here and it is a tempting one, because a bond either exists or it does not and a cutoff radius looks like the way to decide. A chosen cutoff makes the reading report the cutoff. The reading is therefore not whether two atoms are within some chosen distance. It is how far the observed set of distances sits from the set a random packing of the same atoms would give, with nothing chosen.

4.2 ORDER IS NOT AUTHORSHIP

A departure from the disordered background does not measure human production. Unauthored sequences depart: the gaps between primes return a departure outside the band every memoryless process occupies, and nobody wrote the primes.

A crystal is the same refutation in matter. A single-element metal packs into a lattice of long-range order, its atoms on a repeating grid recoverable to one voxel, and no person arranged it. The order is real, the departure is large, and the cause is physics minimizing an energy instead of an author. Chemistry belongs with the primes and not with the prose: it is a domain where the arrangement is far from disordered and nothing produced it deliberately. A reading that took the departure as evidence of design would be wrong here on every mineral in the ground. The departure measures arrangement, and arrangement is not authorship, and the crystal is the cleanest place that difference can be shown.

4.3 WHICH SHUFFLE, AND WHAT IT ISOLATES

The shuffles are graded, and the grades transfer. A full permutation of the atoms destroys every arrangement at once and says only that arrangement is present. It does not say at what scale. A shuffle that keeps local neighborhoods and destroys long ones separates a bond, which is local, from a lattice, which is not: the bond survives a long-range shuffle and the lattice does not. A sweep of the shuffle's reach locates where a structure lives. Scrambling within neighborhoods and keeping the coarse layout does the reverse, and separates the packing of a phase from the identity of the atoms packed.

None of these needs a distribution assumed for the disordered state. A model of a random packing could be wrong. The atoms of the actual substance, with their positions deleted, cannot be, because it is the substance with one property removed and not a guess about what the substance is like. That is the sense in which no set here is bounded and no null here is derived: the null is the data, shuffled, and the only quantity reported is the distance to it.

5

WHAT A MOLECULAR READING WOULD FIND

Nothing in this chapter is measured here. It is the design chemistry adds to the method and the predictions that design makes, stated before the reading exists. The method reads a protein and a crystal but has no representation for a molecule itself, and no bond-length test oracle is compiled.

5.1 WHAT A MOLECULAR READING NEEDS

Two parts are new and the rest is reuse, and one thing chemistry needs belongs to another subject and is only consumed here. The element itself, its proton count Z, its electron set, the Pauli exclusion [5] behind the shell counts, and the periodic recurrence, is the atomic structure, and it is stated once and shared by every subject that reads it. A second element table under chemistry would be a second source of truth for a fact chemistry does not establish.

What chemistry adds is the bonding layer. A molecule needs a representation, atoms as points carrying element vectors and bonds as the vectors between them, beside the protein and crystal representations, since the three share what an atom is. The bond lengths need a test oracle holding the tabulated lengths as facts, apart from the language family trees. Nothing else is added. The octet runs on the screening's existing intersection, the disordered packing on the existing shuffles, and the departure on the existing measures. If the reuse holds, no primitive learns that chemistry exists.

5.2 THE PREDICTIONS

First. The octet as a necessary condition will refuse no real molecule and will prune candidate arrangements, with the error one directional. A structure the octet admits may still be the wrong isomer. A verification pass stays mandatory, and the count of survivors will move with the composition exactly as the lattice cascade's does: a formula holding one element will over-admit because every atom is compositionally alike, and a formula holding several will prune hard.

Second. A bond-length test oracle will be a positive control on par with the crystal cell edge and stronger than any family tree, because valence fixes the value and a substance cannot argue with it. A reading of molecular geometry checked against it will not be able to hide an error inside a tolerance, for the same reason the backbone period can be called wrong the moment it leaves three.

Third. A crystal of a single element will return a large departure from the disordered packing with nothing having authored it, placing chemistry beside the prime gaps and outside the band every memoryless process occupies. A reading that took departure as evidence of production will be wrong on every mineral, and this is the cleanest place to show that arrangement is not authorship.

Fourth. Reading a periodic property along Z will recover the shell structure only if the partition it is read against comes from outside the sequence. The method reads a recurrence in two composed steps, and both fix a single period. The period estimate scores a candidate period against its own family of multiples, and a repeat at three is read as three and not as its harmonic six. The periodic reference then builds the maximum-entropy background at that one period, binding each position to the positions congruent to it modulo P. A periodic property along Z is not one period: the shell lengths are $2, 8, 8, 18, 18, 32$, a sequence that grows. No single P makes the phase classes fall on the shell closures, and a reading restricted to one period reports a harmonic, a single wrong period, or nothing.

A partition is stipulated, estimated by a sweep, or supervised from outside. It is not stipulated here, since a judgment-picked boundary set is the bounding this research paper refuses. It cannot be estimated, since a sweep for one period cannot recover a growing one, and that attempt is the trap. It is supervised: the shell boundaries are ground truth the atomic structure carries, the same kind of fact as the bond-length test oracle and reached from a second direction. The prediction is that the real assignment departs far from its shuffle once the boundaries are supplied, and sits inside the shuffle band to a single-period reading without them. On synthetic sequences, holding no element data, a single-period reading departs on a constant recurrence and goes flat on a growing one, while a reading handed the boundaries departs. This is the harmonic trap for one period, arriving again where the recurrence grows.

Fifth. No reading here will need a bounded element set or a chosen cutoff radius, and any that appears to will be measuring the bound. The bound will have to come off, and the reading will improve when it does.

6

REFERENCE VALUES AND AVAILABILITY

The specific magnitudes this research paper quotes are standard reference values: the covalent and van der Waals radii, the Pauling electronegativities, and the bond lengths, including the backbone values near 1.46, 1.52 and 1.33 angstroms. They are recorded in crystallographic and gas-phase spectroscopic tabulations and in the IUPAC recommendations. Each is quoted to the precision of *near* and is not cited to a specific table here; a value used in a measurement needs the table it came from beside it.

The method, the protein and crystal representations and the screening are in the orior repository, `https://github.com/dstroy0/orior`.

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